



Mohit
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B.Tech.
Gender: Male
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Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	8.03
Intermediate	RBSE	Noble Public School Deolawas	2021	99.40%
Matriculation	RBSE	S.P.S. Sen.Sec.School,Khandwa	2019	94.50%

Pursuing a **Dual Minor** in **Computer Science & Engineering** and **Artificial Intelligence & Data Science**

SCHOLASTIC ACHIEVEMENTS

- Received the **Inspire Award** for being in top **1** percent among **0.25 million** students by **RBSE** [Aug'21]
- Secured a position in the top **3%** in the JEE Advanced examination among **0.17** million aspirants [Sept'22]
- Ranked in the top **0.5** percent among **1 million** candidates appeared in JEE Main examination [July'22]
- Secured **3 AAs** (highest grade point) in various academic courses for exceptional performance [July'23]

PROFESSIONAL EXPERIENCES

Samvid.AI | *SDE Intern* [May'24 - July'24]

Awarded **Letter of Recommendation** from the company's CEO for exceptional performance & work ethics

- Developed an AI chatbot integratable with various **LLMs**, including **GPT, Azure, Ollama**, and **Mistral**
- Created a **Summarization** API using OpenAI's GPT-4 from uploaded files and Containerized using **Docker**
- Implemented Authentication using **Google OAuth** for secure Login and Signup using **Supabase** and **Next.js**
- Integrated **Role-Based Access Control** with **multi-tenancy** by linking users with roles using **TypeScript**

Qrata | *Machine Learning Intern* [Feb'24 - Apr'24]

- Leveraged **LangChain & GPT** to generate resume embeddings & insights, increased user engagement by **40%**
- Used **multi-threading** to fetch files, boosting efficiency by **80%**. Created a **cron scheduler** for automation.
- Performed **K-Means** clustering to build matching system, created smart search bar querying from **VectorDB**
- Built **REST APIs** with **Node.js & MongoDB**, automated filtering of **15k+** talents using semantic extraction

KEY TECHNICAL PROJECTS

Speakspear AI Chatbot | *WiDS | Analytics Club, IIT Bombay* [Dec'23 - Jan'24]

- Built a **GPT Bigram Model** with **1M** parameters using **PyTorch** trained on Shakespeare's writings
- Converted Shakespeare's texts into vectors using **Frequency encoding** with further **Positional encoding**
- Utilized **multi-head self-attention blocks** within **Transformer architecture** for better understanding

Alumination Web Portal | *Student Alumni Relations Cell | IIT Bombay* [Aug'23 - Oct'23]

- Built Alumination website, which provided a seamless platform for organizing a 2-days fest with **10k+** users
- Developed **React** web-app using **Styled-components** and implemented **Horizontal Scrolling, Parallax Animation, 3D carousel, React-Bootstrap**, Count-up Counter, **Django Rest APIs** with **MySQL**
- Integrated the Add to Calendar **API** for sending **auto-generated** personalized mail as an event reminder

Anime Face Generator | *HelloFOSS | Web and Coding Club* [Oct'23 - Dec'23]

- Developed Anime Image Generator using Generative Adversarial Networks, employing **PyTorch** on a **GPU**
- Utilized a **CNN** for Discriminator & **Transposed Convolution** layer for Generator to create realistic anime
- Trained **GAN** model on Anime Face Dataset of **63k+** images, handling GANs & optimizing hyperparameters

SARC Centralized Website | *Centralized Web application* [Aug'23 - Oct'23]

- Streamlined registrations across **10+** web portals by creating a centralized website using **ReactJS** and **Django**
- Integrated **API** endpoints from other portals for seamless and **automated** population of registration data
- Implemented secure authentication using **Django's** built-in authentication system after testing by **Postman**
- Developed **data mapping** across portals & used advanced libraries like **VantaJS** and **GSAP** for revamping.

Predictioneer | *Time Series Analysis* | *AZeotropy, IIT Bombay* [Dec'23 - Jan'24]

- Pre-processed data using **Imputations** and **log-transformations** & effectively performed **ADFuller** test.
- Utilized a deep **LSTM** model to **forecast** water level depth, leveraging historical data for accurate predictions.
- Implemented **ARIMA** and **SARIMA** models for water level data, providing forecasts over the next 3 years

Competitive Programming | *Season of Code* | *Web and Coding Club, IIT Bombay* [May'24 - Present]

- Solved **400+** problems of **DP**, **graphs**, hashing, STL, Trees, Heap & achieved **1100+** **Codeforces** Rating
- Implemented advance algorithms like **Knapsack**, Dijkstra's, Prim's, **Kruskal's**, KMP algorithm, BFS, DFS

Conversation Summarizer | *Course Project* | *Prof. Asim Tewari* [Aug'23 - Dec'23]

- Finetuned **Tiny-Llama-1.1B** model to enhance its conversation summarization ability using Instruction-based **Parameter Efficient Fine-Tuning (PEFT)**, reducing computation expense and memory usage
- Further reduced memory usage and computation by utilizing the **qLoRA** technique for **4-bit quantization**
- Achieved strong performance with **transformer-based PyTorch** model, trained on only **0.1%** of parameters

Optimize Stock Trading Strategies | *Finance Club, IIT Bombay* [June'24 - Present]

- Solved Inverted Pendulum problem using **Deep Q-Network** and **Deep Deterministic Policy Gradient**
- Working on preprocess **NIFTY50** stock data, training and fine-tuning **RL** models for optimal performance
- Testing model performance against models, including **ARIMA** and **LSTM**, to optimize trading strategies

SARCasm | *Student Alumni Relations Cell* | *IIT Bombay* [Dec'23 - Jan'24]

- Developed a gamified crypt-hunt **quizzing platform** using **django** framework with **2000+** participation
- Incorporated interesting features like **time-scheduled** bonus questions, live leaderboard, risk-reward tradeoff

POSITIONS OF RESPONSIBILITY

Web Coordinator | *Student Alumni Relations Cell* [June'23 - May'24]

Worked in a team of 80+ organizing 50+ events to foster relations in 60k+ alumni and 15k+ students

Technical	• Built a Django-based web app & algorithm for greying out mentors during Alumni Student Mentorship Program registration, effectively integrating data from 5k+ previous mentors • Developed personalized yearbooks for 500+ users with code-based React-pdf renderer
Initiatives	• Implementing a threaded messaging feature facilitating 3k+ alumni-student engagement • Contributed to conduction of a technical Hackathon to explore cutting-edge technologies

Seasons of Code Mentor | *Web & Coding Club, IIT Bombay* [May'24 - Present]

- Mentored **8** students for object detection with YOLO, image classification & implementing paper on **GAN**
- Provided resources & assignments on Neural Nets and DeepLearning, and Supported by addressing doubts

TECHNICAL SKILLS

Programming C++, Python, Javascript, MySQL, TypeScript, Flutter, Tailwind, Dart, Jupiter, XCode
DevOps ReactJS, NodeJS, Django, ExpressJS,GetX, Bootstrap, Docker, Postman, Github
Data Science PyTorch, NumPy, Pandas, OpenCV, Matplotlib, TensorFlow, MATLAB, SciKit-Learn

COURSES UNDERTAKEN

Computer Science Data Structures & Algorithms, Computer Programming, Logic for Computer Science, Intro to ML, AI & Data Science, ML & Data Mining, Neural Networks & Deep Learning
Mathematics Calculus, Linear Algebra, Differential Equation, Introduction to Numerical Analysis

EXTRACURRICULARS

Competitions	• Secured Ace and Elite Coder award for 1st position in Coding Ninja DSA Contest [’23] • Participated in Quant Quest AlgoTrading Competition with 400+ participation [’23]
Technical	• Mentored 10+ students for age & gender detection with OpenCV & Feature Detection [’23] • Led 2 teams-of-4 in CodeWars, a python-based game conducted by Web & Coding Club [’23]
Miscellaneous	• Runner up in NCC Volleyball General Championship with 300+ participation [’23] • Achieved Noble Award for contribution to rural education by Noble Education Group (’23)